SmartHire AI: Candidate Discovery & Ranking

AI-Powered Talent Matching for Senior AI Engineer (Founding Team)

Redrob Data & AI Challenge Solution | Technical Overview & Architecture

1. Executive Summary & The Recruitment Challenge

Standard recruitment tools fail because they focus on keyword matching rather than role understanding. SmartHire AI acts like an expert recruiter, utilizing semantic similarity and multi-dimensional behavioral signals to surface candidates who genuinely fit the Founding Team role at Redrob.

- • **Beyond Keywords:** Matches structural competencies (e.g., building recommendation systems) rather than simple keyword matches (e.g., 'Pinecone' or 'RAG' tags).
- • **The Founding Team Bar:** Requires deep ML depth (dense retrieval, evaluation metrics) combined with a scrappy, product-focused shipping attitude.
- • **Intent & Availability:** Incorporates actual platform engagement, notice periods, and location alignments into the ranking criteria.
- • **Honeypot Safeguards:** Automatically detects and disqualifies simulated/invalid profiles to ensure high shortlist reliability.

2. System Architecture & Processing Pipeline

A hybrid, precomputation-based retrieval and scoring architecture designed for maximum performance.

Offline Precomputation

Pre-computes candidate embeddings from full profiles (Current Title, Headline, Experience, Summary, Skills, History, Education) using SentenceTransformers all-MiniLM-L6-v2. Restricts token footprint and saves to Disk.

Semantic Matching

Loads job description from DOCX and encodes it on demand. Computes matrix dot product of normalized embeddings for lightning-fast (under 5s) candidate-JD cosine similarity.

Hybrid Multi-Factor Scorer

Calculates composite scores by combining semantic similarity (60%) and explicit skill alignment (40%), and then applies multipliers based on experience, location, and behavior.

3. Profile Quality & Honeypot Detection

To protect recruiters from spam, the scoring engine scans candidates for logical contradictions and disqualifies them immediately.

- • **Startup Foundation Conflict:** Detects candidates claiming to work at startups prior to their official founding date (e.g. Swiggy in 2010 when founded in 2014). Verified against a curated truth table of 30+ startups.
- • **Experience-Timeline Discrepancy:** Checks if stated years of experience exceed the actual timeline span of their career history (earliest start date to 2026) by more than 1.5 years.
- • **Skill Duration Anomalies:** Flags profiles containing multiple 'expert' or 'advanced' skills with 0 months of duration (indicates randomized or spam keyword stuffing).
- • **Strict Exclusions:** Automatically sets the score of any candidate triggering these checks to 0.0, ensuring 0% honeypot rate in the top 100 shortlist.

4. Multi-Factor Scoring Formulation

Our ranking system applies a transparent, explainable formula to grade candidates:

```
Final Score = ( [0.6 * Semantic Similarity + 0.4 * Skill Match] * Experience
Multiplier * Behavioral Multiplier ) + Activity Bonus
```

- • **Semantic Similarity (60% weight):** Cosine similarity between candidate profile embedding and the JD text.
- • **Skill Match Score (40% weight):** Evaluates required skills (Embeddings/RAG, Vector DBs, Python, Eval Frameworks) and preferred skills (Fine-tuning, LTR) using proficiency and duration metrics.
- • **Experience Multiplier:** Aligns candidate experience with the target 5-9 year curve, applying a penalty for under-experience and a slow decay for over-qualification.
- • **Behavioral Multiplier:** Multiplicative discount based on active signals (recruiter response, notice period, location willingness).

5. Behavioral Signals & Availability Modifiers

A candidate who is technically perfect but unavailable or unresponsive is a poor hire. We calibrate ranks using live indicators:

- • **Recruiter Response Rate:** Provides a 1.05x boost for highly responsive candidates (>=75%), while heavily penalizing unresponsive ones (<10% response rate receives a 0.55x penalty).
- • **Notice Period Multipliers:** Sub-30-day notice period receives a 1.05x speed-joiner boost. Long notice periods (90+ days) receive a 0.65x penalty to favor fast hires.
- • **Location Alignment & Relocation:** Local candidates (Pune/Noida/Delhi NCR) get a 1.10x local boost. Tier-1 Indian city candidates willing to relocate get a 1.0x neutral multiplier, while unwilling candidates are penalized (0.70x).
- • **Platform Activity Bonuses:** Small additive bonuses (+0.02 to +0.03) are awarded for candidates with high recruiter saves, profile views, or strong GitHub contribution activity.

6. Offline Optimization & Scalability

Meeting strict online processing constraints: <5 mins execution and <16 GB RAM.

- • **Precomputed Sentence Embeddings:** Candidates are embedded offline using the lightweight, high-performance 'all-MiniLM-L6-v2' model. Embeddings are stored as a 153MB dense matrix (`candidate_embeddings.npy`) on disk.
- • **High-Speed Vectorized Matching:** During online execution, the Job Description is embedded in milliseconds, and semantic similarities across all 100k candidates are calculated in under 0.1 seconds using vectorized matrix multiplication.
- • **Single-Pass Scorer Execution:** Candidate metadata loading and custom scorer evaluation are executed in a single-pass loop, completing the entire ranking pipeline in under 5 seconds.
- • **Zero API Dependencies:** The system runs entirely offline with local SentenceTransformers weights and logic, ensuring absolute compliance with challenge guidelines.

7. Interactive Recruiter Dashboard

A premium recruitment workspace designed to make decisions faster and clearer.

- • **Glassmorphic Dark Design:** Premium visual style utilizing a curated, cohesive dark mode color scheme (Indigo, Emerald, Charcoal) and Outfit typography.
- • **Interactive Filter Workspace:** Dynamic sidebar allowing recruiters to filter by experience (min-slider), notice period, local Noida/Pune status, and activity intervals.
- • **Score Breakdown & Explainability:** Displays individual candidate sub-scores (semantic, skills, experience, engagement) and lists explicit strengths, weaknesses, and missing skills.
- • **Career Timeline & Skill Gaps:** Visualizes candidates' career timelines dynamically and checks their skills against required/preferred categories with check/cross icons.

8. Verification & Validation Metrics

Rigorous testing to guarantee ranking reliability, consistency, and compliance.

- • **Auto-Validator Alignment:** Verified that candidate rankings contain exactly 100 entries in `submission.csv`, sorted in strictly non-increasing order of scores.
- • **Deterministic Tie-breaking:** Resolves equal scores using candidate ID in ascending alphabetical order, ensuring matching runs produce identical shortlists every time.
- • **0% Honeypot Kurz-Verification:** Verified that honeypot candidates are filtered out of the top 100 recomendations, ensuring zero recruiter-facing quality regressions.
- • **Schema Verification:** Runs the provided `validate_submission.py` to guarantee that output columns ('candidate_id', 'rank', 'score', 'reasoning') conform to requirements.

9. Future Roadmap & Conclusion

Key takeaways and long-term features for SmartHire AI:

- • **Summary:** SmartHire AI successfully bridges the gap between structured signals and semantic context, identifying the absolute best fits from 100k records in seconds.
- • **Two-Stage LLM Re-ranking:** For future scale, we can feed the top 500 candidates into a local, quantized LLM (e.g. Llama-3-8B-Instruct) to perform advanced instruction-based re-ranking.
- • **Fine-tuned Embeddings:** Contrastive fine-tuning of the embedding model using historical hire/rejection datasets can align semantic embeddings closer to actual company-specific preferences.
- • **Hybrid Search Indexing:** Implementing Qdrant or Milvus in the backend to enable vector and lexical hybrid search dynamically as new candidates are added.